

Lab 1 — Requirements Engineering & UML Use-Case Modelling

Problem Statement #39 — Second-Hand Product Escrow Marketplace

Student: Dhanush R | SRN: PES1UG24AM427

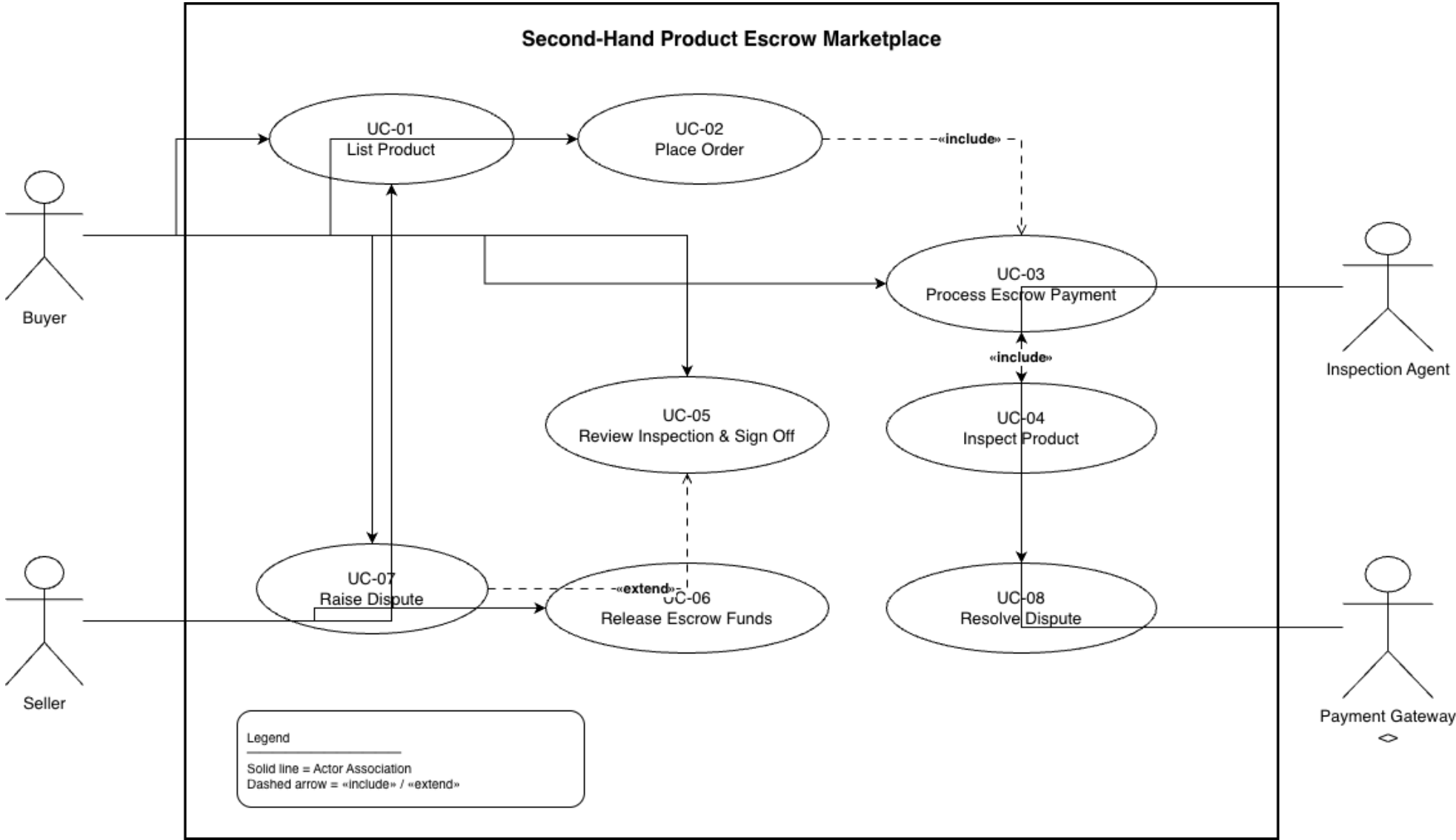
Requirements Table — Second-Hand Product Escrow Marketplace PES1UG24AM427 Dhanush R Problem #39					
	None	None	None	None	None
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall lock the buyer's payment in escrow and transition the order status to "Under Inspection" until the Inspection Agent submits physical verification sign-off.	High	1. Given a buyer completes checkout, the payment amount is deducted from the buyer's account and held in an escrow ledger entry within 5 seconds. 2. The order status field is set to exactly "Under Inspection" immediately after the escrow lock is confirmed. 3. The escrowed funds cannot be credited to the seller account until the Inspection Agent submits a verification sign-off; any attempt to release before sign-off returns an error code <code>ESCROW_NOT_CLEARED</code>.	Holding funds in escrow before inspection protects the buyer from paying for a product that has not been independently verified, which is the core trust mechanism of the marketplace.
FR-002	Functional	The system shall allow the Inspection Agent to record the physical condition of the product and submit an inspection verification report for the order.	High	1. An authenticated Inspection Agent can submit a report containing at minimum: condition rating (one of: Excellent / Good / Fair / Poor), description text (1–500 chars), and up to 10 photo attachments. 2. On successful submission, the order record stores the report with the agent's authenticated user ID and a UTC timestamp accurate to within 1 second. 3. A report cannot be submitted for an order whose status is not	Independent physical inspection by a certified agent provides objective evidence of product condition, reducing disputes and increasing buyer confidence in second-hand goods.

				"Under Inspection"; such an attempt must return error code <code>INVALID_ORDER_STATE</code> .	
FR-003	Functional	The system shall allow the buyer to review the inspection result and provide a sign-off indicating whether the product condition is accepted.	High	1. After an inspection report is submitted, the buyer receives an in-app notification within 60 seconds and can view the full inspection report including condition rating, description, and photo attachments. 2. The buyer can select exactly one of two actions: "Accept" or "Raise Dispute"; no other state transitions are permitted at this step. 3. Selecting "Accept" records the buyer's authenticated sign-off (user ID + UTC timestamp) against the order and transitions the order status to "Awaiting Release".	Giving the buyer final sign-off authority over the inspection result ensures the buyer retains control over whether their payment is released, upholding the escrow guarantee.
FR-004	Functional	The system shall release the escrowed payment to the seller only when the required inspection verification and buyer sign-off have both been completed.	High	1. The release workflow checks two boolean flags: <code>inspection_verified = true</code> AND <code>buyer_signed_off = true</code>; if either flag is false, the release is rejected with error code <code>RELEASE_CONDITIONS_NOT_MET</code>. 2. When both conditions are true, the escrowed amount is credited to the seller's account within 60 seconds of the release being triggered. 3. After a successful release, the order status is updated to "Completed" and no further modifications to the escrow entry are permitted.	Releasing funds only after dual confirmation (agent + buyer) prevents premature or fraudulent disbursement and is the contractual fulfilment of the escrow agreement.
FR-005	Functional	The system shall allow the buyer to raise a dispute for an order and freeze the associated escrow funds while the dispute is under investigation.	High	1. A buyer can raise a dispute only when the order status is "Under Inspection" or "Awaiting Release"; attempts from any other status must return error code <code>DISPUTE_NOT_ALLOWED</code>. 2. Upon dispute creation, the	A dispute mechanism with immediate fund freeze ensures buyers can contest unsatisfactory inspections without risk of funds being released to

				escrow funds are frozen within 5 seconds, and all release and withdrawal requests against those funds are rejected with error code FUNDS_FROZEN until the dispute is resolved. 3. A dispute ticket is created with a unique ID, the buyer's user ID, the order ID, a UTC timestamp, and an initial status of "Open".	the seller before the issue is resolved.
NFR-001	Non-Functional	The system shall freeze escrow funds and prevent withdrawal of the disputed amount immediately when a dispute is registered.	High	1. From the moment a dispute is registered, any seller withdrawal or escrow release request for the disputed order must be rejected within 500 milliseconds with error code FUNDS_FROZEN under a simulated load of up to 500 concurrent users. 2. The fund-freeze event is written to the audit log with a UTC timestamp before the dispute creation API response is returned to the client, with zero exceptions tolerated over 10,000 simulated dispute registrations in performance testing.	Immediate fund freezing is a safety constraint: any window between dispute registration and freeze could allow a malicious seller to withdraw funds, defeating the purpose of escrow.
NFR-002	Non-Functional	The system shall ensure that escrow transactions, inspection reports, buyer sign-offs, and dispute actions are recorded with authenticated user identity and cannot be modified without authorization.	High	1. Every write operation on escrow transactions, inspection reports, sign-offs, and dispute records must store the authenticated user's ID and a UTC timestamp; records missing either field must be rejected at the persistence layer. 2. Any attempt to modify or delete an existing escrow transaction, inspection report, or sign-off record by a user who did not create it, or who lacks an explicit admin role, must return HTTP 403 Forbidden and be logged in the audit trail. 3. Audit log records themselves	Immutable, identity-linked records provide legal-grade evidence of all marketplace actions, deter fraud, and satisfy regulatory audit requirements for financial transactions.

				must be append-only; an automated integrity check run every 24 hours must report zero tampered or deleted audit entries.	
--	--	--	--	--	--

Exactly 5 Functional Requirements (FR-001 to FR-005) and 2 Non-Functional Requirements (NFR-001 to NFR-002).



USE-CASE FLOW SPECIFICATION

Use Case: UC-05 — Review Inspection & Sign Off

Actors: Buyer, Second-Hand Product Escrow Marketplace

PRECONDITIONS

- The purchaser maintains a verified, active transaction for a pre-owned item.
- Escrow protocols have locked the buyer's capital, and the transaction is current in the "Under Inspection" status.
- An authenticated Inspection Agent has officially submitted the physical verification report to the ledger.

POSTCONDITIONS

- Upon buyer acceptance, an authenticated sign-off is logged, transitioning the order to a state eligible for fund disbursement.
- Should the buyer reject the condition, a dispute is initiated, ensuring escrowed assets remain frozen and protected.

MAIN SUCCESS SCENARIO

1. The Buyer accesses the specific order currently awaiting physical verification confirmation.
2. The System renders the comprehensive inspection report, including condition ratings and independent verification details.

3. The Buyer performs a thorough review of the agent's recorded inspection findings.
4. The Buyer triggers the authenticated action to "Accept" the product based on the report.
5. The System persists the buyer's digital sign-off (User ID + UTC timestamp) within the order record.
6. The System validates that both the inspection sign-off and buyer acceptance flags are set to true.
7. The System updates the order status to "Awaiting Release," permitting the disbursement of funds to the seller.
8. The use case terminates with a successful state transition.

ALTERNATE FLOW — BUYER DISPUTES INSPECTION

- 4A1. At the decision point, the buyer identifies that the verified product condition fails to meet expectations.
- 4A2. The Buyer selects "Raise Dispute" as the primary action instead of accepting the inspection result.
- 4A3. The System generates a unique dispute ticket and immediately freezes the associated escrowed funds within 5 seconds.
- 4A4. The System logs the dispute event in the audit trail and routes the ticket for internal investigation.
- 4A5. Escrow release is strictly prohibited and returns error code FUNDS_FROZEN until the conflict is resolved.